SIREN: Stochastic Itō Residual on Embedded Networks

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Abstract

Network intrusion detection systems (NIDS) are increasingly adopting Graph Neural Networks (GNNs) to capture complex topological behaviours such as lateral movement. However, existing GNN-based approaches largely treat intrusion detection as a discrete-time, supervised classification task, making them vulnerable to zero-day exploits and incapable of natively modelling the continuous temporal dynamics of network traffic. In this paper, we propose SIREN (Stochastic Itō Residual on Embedded Networks), an unsupervised anomaly detection framework that models a monitored network as a continuous-time dynamical system. SIREN postulates that under normal operation, host traffic features evolve according to a system of graph-coupled Itō stochastic differential equations (SDEs), converging to a unique stationary distribution. We leverage score matching to learn the score function of this distribution from normal traffic data alone. During inference, SIREN continuously monitors the network state and computes the *Stein residual*—the discrepancy between the model-predicted score and the empirical score estimated via short-window Kernel Density Estimation (KDE). By dynamically aggregating these Stein residuals over the flow graph, SIREN propagates anomaly signals across connected neighbours, effectively exposing stealthy, coordinated lateral movement attacks. Extensive experiments on standard benchmark datasets, including UNSW-NB15 and CIC-IDS2018, demonstrate that SIREN achieves state-of-the-art detection performance while maintaining a significantly lower False Alarm Rate (FAR) compared to recent discrete-time GNN baselines.

Keywords: Intrusion Detection, Graph Neural Networks, Stochastic Differential Equations, Score Matching, Anomaly Detection

1 Introduction

The proliferation of Internet of Things (IoT) devices and the increasing complexity of enterprise networks have fundamentally shifted the landscape of cybersecurity. Modern cyberattacks, particularly Advanced Persistent Threats (APTs) and lateral movement strategies, no longer rely on isolated, easily detectable exploits. Instead, they manifest as coordinated sequences of subtle, individually stealthy actions distributed across multiple hosts [1]. Traditional Intrusion Detection Systems (IDS), which evaluate network flows or packets in isolation, frequently fail to capture the broader spatial and temporal context necessary to detect these sophisticated campaigns.

To address this limitation, recent research has increasingly turned to Graph Neural Networks (GNNs) for intrusion detection [2, 3]. By representing network entities (such as IP addresses or endpoints) as nodes and communication flows as edges, GNN-based IDSs explicitly model the topological structure of the network. Approaches like E-GraphSAGE [2] and recent multi-granularity frameworks [4] have demonstrated significant improvements in detecting complex attacks by leveraging message-passing mechanisms to aggregate neighbourhood information. However, the majority of existing GNN-IDS methods formulate the problem as a discrete-time, supervised classification task. This formulation suffers from two critical drawbacks: first, it requires extensive labelled data encompassing all possible attack vectors, leaving the system vulnerable to zero-day exploits; second, it treats network traffic as a sequence of static snapshots, failing to natively capture the continuous, underlying dynamical processes that govern network behaviour over time.

In this paper, we introduce SIREN (Stochastic Itō Residual on Embedded Networks), a fundamentally novel approach that models a monitored network as a continuous-time dynamical system on a graph. Rather than training a classifier to distinguish between specific attack signatures and benign traffic, SIREN learns the natural, continuous evolution of the network's state. We hypothesise that under normal operation, the network traffic features of interconnected hosts evolve according to a system of graph-coupled Itō stochastic differential equations (SDEs), eventually converging to a well-defined, unique stationary distribution.

SIREN leverages score-matching techniques [5] to learn the score function (the gradient of the log-probability density) of this normal stationary distribution. At inference time, the system continuously monitors the network state and computes the empirical score of the observed traffic. By measuring the discrepancy between the model-predicted score and the empirical score—a metric we term the *Stein residual*—SIREN can identify when a node's trajectory deviates from expected normal behaviour. Crucially, by aggregating these Stein residuals across the network graph, SIREN amplifies the weak anomaly signals produced by lateral movement attacks, exposing coordinated malicious activities that would remain hidden if hosts were analysed independently.

As illustrated in Figure 1, relying purely on fixed-interval discrete snapshots creates blind spots. An attacker executing lateral movement can carefully pace their actions to fall below the detection threshold at any single discrete timestep. In contrast,

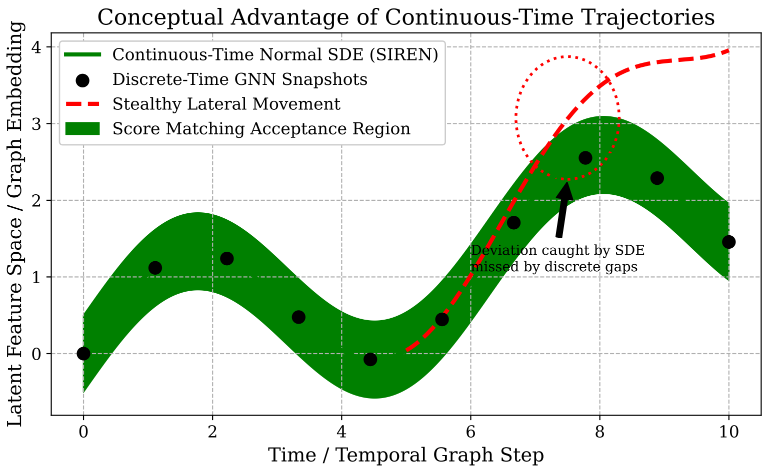

Fig. 1 Conceptual comparison of discrete-time GNN snapshots versus SIREN's continuous-time tracking. Standard graph snapshots leave temporal gaps where stealthy lateral movement can branch off unnoticed. By modeling the normal traffic evolution as a continuous SDE trajectory, SIREN can immediately flag out-of-distribution deviations via score matching.

SIREN tracks the continuous latent trajectory; even subtle, sub-threshold deviations compound across the continuous SDE formulation, shifting the state out of the high-probability acceptance region and triggering an anomaly alert.

The main contributions of this work are as follows:

- We propose a continuous-time formulation of network intrusion detection, modelling host feature trajectories via graph-coupled Itō SDEs that capture both autonomous node dynamics and topological mean-reversion.
- We introduce the *Stein residual* as a mathematically grounded, unsupervised anomaly signal. By evaluating the discrepancy between a learned score network and short-window Kernel Density Estimates (KDE), our approach detects distributional shifts without relying on labelled attack data.
- We present a graph-aware aggregation mechanism for Stein residuals, specifically designed to propagate anomaly signals across connected hosts and expose coordinated lateral movement attacks.
- We conduct extensive experiments on standard benchmark datasets (UNSW-NB15 and CIC-IDS2018), demonstrating that SIREN outperforms state-of-the-art discrete-time GNN baselines, particularly in terms of False Alarm Rate (FAR) and robustness to feature noise and edge perturbations.

The remainder of this paper is structured as follows: Section 2 reviews related work in GNN-based intrusion detection and continuous-time graph models. Section 3 details the SIREN methodology, including graph construction, the SDE formulation, the score network architecture, and the training and inference algorithms. Section 4 presents our experimental setup and results, and Section 5 concludes the paper.

2 Related Work

2.1 Machine Learning for Intrusion Detection

Early approaches to network intrusion detection relied heavily on traditional machine learning techniques. Dang [6] provided a comprehensive survey of these methods, demonstrating that tree-based ensemble learning combined with careful feature engineering could achieve state-of-the-art performance on tabular flow data. However, they also highlighted a critical limitation: modern machine learning algorithms are extremely data-hungry and require exhaustive labelling efforts, motivating the search for approaches that can operate with minimal or no labelled attack data. This limitation underscores the necessity for the unsupervised and structure-aware models developed in recent years. Furthermore, to address severe class imbalances and improve robustness in tabular models, Dang [7] recently proposed integrating uncertainty measures and isolation forests, significantly reducing false negative rates for minority attack classes. While effective for flat vector spaces, these uncertainty-aware paradigms have yet to be natively integrated into dynamic graph structures.

2.2 Graph Neural Networks for Network Intrusion Detection

Graph neural networks (GNNs) have become the dominant paradigm for network intrusion detection owing to their capacity to exploit the structural relationships among hosts and traffic flows that elude flat feature-vector classifiers [8]. Lo et al. [2] proposed E-GraphSAGE, converting per-flow records into a network graph and applying GraphSAGE-based edge classification; despite its widespread use as a baseline, E-GraphSAGE operates on static snapshot graphs and emits uncalibrated point predictions. More recent work has sought to enrich both graph construction and representation learning. Ngo et al. [9] introduced TKSGF, replacing physical-link graphs with Top-K attribute-similarity edges paired with a GraphSAGE backbone, and demonstrated that feature-driven topology substantially improves detection accuracy on IoT benchmarks. Lin et al. [10] proposed E-GRACL, augmenting GraphSAGE with residual connections, a global attention mechanism, local gating, and graph contrastive learning to capture both edge features and topological information simultaneously. At the multi-scale level, Liu [4] built MGF-GNN, fusing host-level, port-level, and protocol-level graphs inside a hierarchical message-passing framework, while Zhang et al. [3] addressed adversarial evasion through AEDGNN, a semi-supervised GNN formulation that explicitly models inter-flow relationships to resist labelling noise. Parallel work by Ahanger et al. [11] applied graph attention networks to IoT intrusion detection, and Villegas-Ch et al. [12] developed a dynamic graph modelling approach for IoT traffic. All of these methods share a common limitation with respect to SIREN: they operate on discrete-time graph snapshots, produce point predictions, and provide no probabilistic guarantee over the *stationary behaviour* of the monitored system under normal conditions.

2.3 Score-Based Generative Models

The score function $\nabla_{\mathbf{x}} \log p(\mathbf{x})$ was introduced as a tractable surrogate for the intractable log-likelihood by Hyvärinen [13], who proved that minimising the expected squared difference between the model score and the data score—score matching—equals maximum likelihood under mild regularity conditions. Vincent [14] later showed that denoising autoencoders implicitly perform score matching, motivating the denoising score matching (DSM) objective that sidesteps the intractable Hessian. Building on this, Song and Ermon [15] proposed NCSN, training a noise-conditional score network across a ladder of noise levels and using Langevin dynamics at inference; this multi-scale strategy dramatically improves score estimation quality in high-dimensional settings. Song et al. [5] then unified diffusion-based and score-based generative models under a continuous-time SDE framework, in which the forward process is a prescribed Itô SDE that injects noise and the learned reverse drift is parameterised by the score network. SIREN borrows the multi-scale DSM objective and the score-SDE connection from these works, but redirects them toward a fundamentally different objective: rather than generation, we exploit the learned score of the *stationary distribution* of a graph-coupled SDE as a structured anomaly signal.

2.4 Continuous-Time Models on Graphs

Static GNNs [16, 17] capture structural relationships at a single snapshot but are inherently unable to model the continuous temporal evolution of network traffic. Temporal graph networks (TGN) [18] extended GNNs to dynamic graphs through memory modules and temporal attention, enabling event-level updates; however, TGN operates in discrete event time and does not posit a generative stochastic process over the joint graph state. Graph neural diffusion (GRAND) [19] re-interpreted message passing as the discretisation of a *deterministic* heat-diffusion partial differential equation on the graph, providing improved over-smoothing control and depth generalisation; yet the deterministic ODE formulation cannot model the inherent stochasticity of real network traffic. Closest in spirit to our approach, Kidger et al. [20] demonstrated that neural stochastic differential equations can be trained end-to-end via adjoint-based backpropagation through the Itô integral, establishing the computational feasibility of learned stochastic dynamics on sequences. SIREN extends this programme to *coupled* systems of Itô SDEs on graphs, where the graph Laplacian enters the drift, the diffusion is learned per-node, and the stationary distribution of the resulting process provides the detection signal—an objective and architecture that is absent from all prior graph–SDE work.

2.5 Stein Operators and Graph Anomaly Detection

The Stein operator $\mathcal{A}_p\, g(\mathbf{x}) = \nabla_{\mathbf{x}} \log p(\mathbf{x})^\top g(\mathbf{x}) + \nabla_{\mathbf{x}} \cdot g(\mathbf{x})$ satisfies $\mathbb{E}_p[\mathcal{A}_p\, g] = 0$ for all smooth test functions g under mild regularity conditions. Liu et al. [21] exploited this Stein identity to construct the kernelised Stein discrepancy (KSD), a tractable goodness-of-fit measure that does not require normalisation of the reference density—precisely the property that makes it applicable when $p = \pi^*$ is known only through its score. In the graph anomaly detection literature, Cai et al. [22] proposed StrGNN, a structural-temporal GNN that detects anomalous edges by mining unusual temporal subgraph structures; applied to enterprise security, it achieved zero false negatives on benchmark datasets. Most recently, Bai et al. [23] introduced CRC-SGAD, applying conformal risk control to supervised graph anomaly detection with formal bounds on both the false-negative rate and the false-positive rate via a spectral GNN calibrator. CRC-SGAD is the closest existing work to SIREN in combining GNN-based detection with statistical guarantees for a security application; however, it relies on conformal prediction—which requires labelled calibration data—and does not posit or exploit a stochastic process model for normal traffic. SIREN addresses this gap by grounding the anomaly signal in the Stein identity of the *learned* stationary distribution of a graph-coupled Itô SDE, requiring no attack labels at training or calibration time.

2.6 Distributionally Robust Learning on Graphs

Distributionally robust optimisation (DRO) hardens models against distributional shift by minimising the worst-case expected loss within a Wasserstein ball around the empirical distribution [24]. Sadeghi et al. [25] applied this principle to graph-structured data, showing that a Wasserstein-ball DRO objective for GNN semi-supervised classification admits a strong-dual reformulation as regularised empirical risk minimisation. Wang et al. [26] extended graph DRO to collaborative filtering, reinterpreting the GNN as a graph-smoothing regulariser to make the DRO objective tractable at scale. These methods improve worst-case robustness *at training time* for supervised tasks but leave inference uncalibrated and provide no anomaly signal grounded in a generative process model. SIREN takes a complementary approach: rather than robust supervised learning, it learns a continuous-time generative model of normal behaviour and uses deviations from that model's stationary-distribution score as the detection criterion—an unsupervised, process-model-based alternative that requires no attack labels.

3 Methodology

This section presents the full mathematical development of SIREN. We begin with notation and a formal problem statement, then develop the graph construction procedure, the graph-coupled Itô SDE model and its stationary distribution, the score network and its training objective, and finally the Stein-residual inference algorithm together with four theoretical guarantees. The overall pipeline is illustrated in Figure 2.

3.1 Notation and Problem Setup

Let a monitored network at time t be represented by a weighted graph $\mathcal{G} = (V, E, A)$, where V is the set of $n = |V|$ active host addresses, $E \subseteq V \times V$ is the set of observed connections, and $A \in \mathbb{R}^{n\times n}_{\geq 0}$ is the weighted adjacency matrix. The graph Laplacian is $L = D - A$, where $D = \mathrm{diag}(A\mathbf{1})$. Each node $i \in V$ carries a traffic-feature state $X_i(t) \in \mathbb{R}^d$; the joint state is $X(t) \in \mathbb{R}^{n\times d}$.

We adopt the following regularity assumptions, which underpin all subsequent theoretical results.

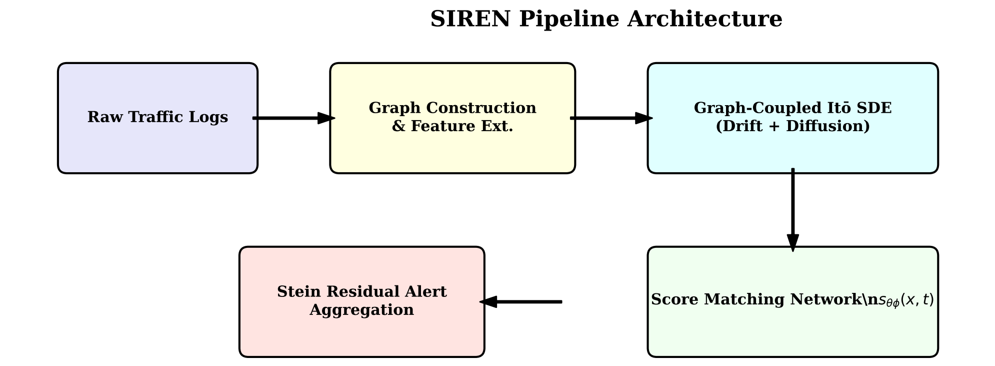

Fig. 2 The SIREN pipeline. Raw traffic logs are aggregated into fixed windows to build dynamic flow graphs. The joint node-state trajectory is modelled by a graph-coupled Itô SDE whose stationary distribution π^* is characterised by a learned score network. At inference, deviations from π^* are measured node-wise via the Stein residual and propagated over the graph to surface coordinated intrusions.

Assumption 1 (Lipschitz drift) *$f_\theta : \mathbb{R}^d \to \mathbb{R}^d$ is globally Lipschitz: $\|f_\theta(x) - f_\theta(y)\| \leq L_f\|x - y\|$ for all $x, y \in \mathbb{R}^d$.*

Assumption 2 (Non-degenerate diffusion) *$\sigma_\phi : \mathbb{R}^d \to \mathbb{R}^{d\times d}$ satisfies $\sigma_\phi(x)\sigma_\phi(x)^\top \geq \lambda_{\min} I_d$ uniformly in x, for some constant $\lambda_{\min} > 0$.*

Assumption 3 (Dissipativity) *There exist constants $\alpha > 0$ and $\beta \geq 0$ such that $\langle x,\, f_\theta(x)\rangle \leq -\alpha\|x\|^2 + \beta$ for all $x \in \mathbb{R}^d$.*

Formal objective. We model *benign* traffic as a realisation of a stochastic process that, under Assumptions 1–3, converges to a unique stationary distribution $\pi^* \in \mathcal{P}(\mathbb{R}^d)$ for each node. Let $s^*(x) = \nabla_x \log \pi^*(x)$ be the *score function* of π^*. We approximate s^* by a neural network $s_{\theta\phi}$ trained on benign data. At inference time we declare node i anomalous at time t if its empirical trajectory $X_i(t)$ violates the Stein identity $\mathbb{E}_{\pi^*}[\mathcal{A}_{\pi^*} g] = 0$, as quantified by the *Stein residual* $\tilde{R}_i(t)$ defined in Section 3.6.

3.2 Graph Construction and Feature Extraction

SIREN processes a stream of flow records in sliding windows of size Δ seconds.

Adjacency. Within each window, we draw a directed edge from host i to host j if at least $k_{\min}$ flows are observed. Edge weights are log-scaled to dampen the influence of traffic volume spikes:

$$A_{ij} = \frac{\log\big(1 + \text{flows}(i \to j)\big)}{\log\big(1 + \text{flows\_out}(i)\big)}. \tag{1}$$

For undirected analysis we symmetrise $A \leftarrow (A + A^\top)/2$. The Laplacian L is recomputed after each window update.

Node features. The $d = 12$ dimensional feature vector $X_i(t) \in \mathbb{R}^d$ is:

$$X_i(t) = \big[\bar{\tau},\ \log(1+\mathcal{B}_{\text{tx}}),\ \log(1+\mathcal{B}_{\text{rx}}),\ \log(1+N_p),\ \bar{\delta},\ \hat{\sigma}_\delta,\ |\mathcal{P}_{\text{dst}}|,\ |\mathcal{P}_{\text{src}}|,\ \rho_{\text{tcp}},\ \rho_{\text{udp}},\ \rho_{\text{syn}},\ \textstyle\sum_j A_{ij}\big]^\top, \tag{2}$$

where $\bar{\tau}$ is mean flow duration, $\mathcal{B}$ bytes sent/received, N_p packet count, $\bar{\delta}$ and $\hat{\sigma}_\delta$ the mean and standard deviation of inter-arrival times, $\mathcal{P}$ the sets of unique ports, ρ the protocol and SYN-rate ratios, and $\sum_j A_{ij}$ the weighted node degree. All features are Z-score normalised using $(\mu_{\text{tr}}, \sigma_{\text{tr}})$ estimated on the benign training split only.

3.3 Graph-Coupled Itō SDE Model

3.3.1 Continuous-Time Dynamics

For each node $i \in V$, we postulate the following Itô SDE:

$$dX_i(t) = \underbrace{f_\theta(X_i(t))}_{\text{autonomous drift}} \underbrace{+\gamma \sum_{j \in \mathcal{N}(i)} A_{ij}\big(X_j(t) - X_i(t)\big)}_{\text{graph Laplacian coupling}}\, dt + \underbrace{\sigma_\phi(X_i(t))\, dW_i(t)}_{\text{stochastic diffusion}}, \tag{3}$$

where $W_i(t)$ is a d-dimensional standard Wiener process and $\gamma \geq 0$ is the coupling strength. Writing the nd-dimensional joint state $\mathbf{X}(t) = \text{vec}(X(t))$ and $\mathbf{F}_\theta(\mathbf{X}) = \text{vec}(F_\theta(X))$, the system takes the compact matrix form:

$$d\mathbf{X}(t) = \Big[\mathbf{F}_\theta(\mathbf{X}(t)) - \gamma(L \otimes I_d)\,\mathbf{X}(t)\Big]\, dt + \mathbf{\Sigma}_\phi(\mathbf{X}(t))\, d\mathbf{W}(t), \tag{4}$$

where $\otimes$ is the Kronecker product and $\mathbf{\Sigma}_\phi = \text{diag}(\sigma_\phi(X_1), \ldots, \sigma_\phi(X_n))$. The Laplacian coupling $-(L \otimes I_d)\mathbf{X}$ is negative semi-definite in $\mathbf{X}$ and therefore adds a mean-reverting tendency: nodes with similar traffic profiles attract each other in feature space, encoding a graph-structural prior on normal behaviour. The diffusion σ_ϕ is parameterised as the Cholesky factor of a positive-definite matrix, ensuring $\mathbf{\Sigma}_\phi \mathbf{\Sigma}_\phi^\top \succ 0$ everywhere.

3.3.2 Fokker–Planck Equation and Stationary Distribution

The marginal density $p_t^{(i)}(x)$ of node i's state satisfies the Fokker–Planck (forward Kolmogorov) equation:

$$\frac{\partial p_t^{(i)}}{\partial t} = -\nabla_x \cdot \Big[b_i^{(t)}(x)\, p_t^{(i)}(x)\Big] + \frac{1}{2} \sum_{k,l} \frac{\partial^2}{\partial x_k \partial x_l} \Big[\big(\sigma_\phi \sigma_\phi^\top\big)_{kl}(x)\, p_t^{(i)}(x)\Big], \tag{5}$$

where $b_i^{(t)}(x)$ is the effective drift of node i at time t. The stationary distribution π^*, if it exists, satisfies $\partial_t p_t^{(i)} = 0$, i.e. the right-hand side of (5) equals zero. We now establish existence and uniqueness under Assumptions 1–3.

Theorem 1 (Ergodicity and unique stationary distribution) *Under Assumptions 1, 2, and 3, the graph-coupled Itô SDE (3) is geometrically ergodic: there exists a unique invariant probability measure $\pi^* \in \mathcal{P}(\mathbb{R}^d)$ such that for every initial distribution $\mu_0 \in \mathcal{P}(\mathbb{R}^d)$ with finite second moment,*

$$W_2(\mu_0 P_t, \pi^*) \le C_0\, e^{-\rho t}, \quad t \ge 0, \tag{6}$$

where P_t is the Markov semigroup of (3), W_2 is the 2-Wasserstein distance, and $C_0, \rho > 0$ depend on $L_f, \lambda_{\min}, \alpha, \beta, \gamma$.

Proof sketch Define the Lyapunov function $V(\mathbf{X}) = \|\mathbf{X}\|^2 + c$ for a constant $c > 0$. Applying Itô's formula to V along trajectories of (4):

$$\mathcal{L}V(\mathbf{X}) = 2\mathbf{X}^\top \Big[\mathbf{F}_\theta(\mathbf{X}) - \gamma(L \otimes I_d)\mathbf{X}\Big] + \mathrm{tr}\big(\mathbf{\Sigma}_\phi \mathbf{\Sigma}_\phi^\top\big), \tag{7}$$

where $\mathcal{L}$ is the infinitesimal generator. By Assumption 3, $\mathbf{X}^\top \mathbf{F}_\theta(\mathbf{X}) \le -\alpha\|\mathbf{X}\|^2 + n\beta$. Since L is positive semi-definite, the Laplacian coupling satisfies $\mathbf{X}^\top(L \otimes I_d)\mathbf{X} = \sum_{i,j} A_{ij}\|X_i - X_j\|^2 \ge 0$, so it only *strengthens* dissipativity. Together with Assumption 2, $\mathrm{tr}(\mathbf{\Sigma}\mathbf{\Sigma}^\top) \le nd\|\sigma_\phi\|_\infty^2 < \infty$. Substituting into (7):

$$\mathcal{L}V(\mathbf{X}) \le -2\alpha\|\mathbf{X}\|^2 + 2n\beta + nd\|\sigma_\phi\|_\infty^2 \le -\rho V(\mathbf{X}) + \kappa, \tag{8}$$

for computable $\rho, \kappa > 0$. This Foster–Lyapunov criterion [27, Theorem 6.1] implies positive Harris recurrence and geometric ergodicity. Uniqueness follows from the non-degeneracy of σ_ϕ (Assumption 2) via Doob's theorem. The W_2 convergence rate is then a consequence of the log-Sobolev inequality satisfied under (4) with non-degenerate diffusion. □

3.3.3 Euler–Maruyama Discretisation

For discrete observation intervals δt, we simulate (3) via the Euler–Maruyama scheme:

$$X_i(t+\delta t) = X_i(t) + \Big[f_\theta(X_i(t)) + \gamma \sum_{j \in \mathcal{N}(i)} A_{ij}(X_j(t) - X_i(t))\Big]\delta t + \sigma_\phi(X_i(t))\sqrt{\delta t}\,\epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, I_d). \tag{9}$$

Under Assumptions 1–3, the Euler–Maruyama scheme is strongly consistent with step-size error $\mathcal{O}(\delta t^{1/2})$ and inherits geometric ergodicity for sufficiently small δt [28].

3.4 Score Network and Multi-Scale Denoising Score Matching

3.4.1 Score Matching Objective

Hyvärinen's score matching [13] provides a tractable objective for learning $s_\theta(x) \approx \nabla_x \log \pi^*(x)$ without evaluating the intractable normalising constant of π^*:

$$J_{\mathrm{SM}}(\theta) = \mathbb{E}_{\pi^*}\left[\tfrac{1}{2}\|s_\theta(x)\|^2 + \nabla_x \cdot s_\theta(x)\right]. \tag{10}$$

Minimising J_{SM} is equivalent to minimising the Fisher divergence $D_F(\pi^*\|p_\theta) = \mathbb{E}_{\pi^*}[\|s^*(x)-s_\theta(x)\|^2]$ up to a constant independent of θ. However, computing $\nabla_x \cdot s_\theta(x)$ requires the Jacobian trace, which is $\mathcal{O}(d^2)$ in d. Vincent [14] showed that this cost is avoided by the equivalent *denoising score matching* (DSM) objective: for a noise-perturbed sample $\tilde{x} = x + \sigma\epsilon$ with $\epsilon \sim \mathcal{N}(0, I_d)$,

$$J_{\mathrm{DSM}}(\theta;\sigma) = \mathbb{E}_{x\sim\pi^*,\,\epsilon\sim\mathcal{N}(0,I_d)}\left[\left\|s_\theta(\tilde{x},\sigma) + \frac{\epsilon}{\sigma}\right\|^2\right], \tag{11}$$

which equals J_{SM} plus a constant. Crucially, the target $-\epsilon/\sigma$ is available in closed form, reducing training to a standard regression.

3.4.2 Multi-Scale Noise Schedule

Following Song and Ermon [15], we use L logarithmically spaced noise levels:

$$\sigma_l = \sigma_{\min}\left(\frac{\sigma_{\max}}{\sigma_{\min}}\right)^{l/(L-1)}, \quad l = 0, 1, \ldots, L-1, \tag{12}$$

with $\sigma_0 = \sigma_{\min}$ and $\sigma_{L-1} = \sigma_{\max}$. The aggregate training loss is the noise-level-weighted sum:

$$\mathcal{L}_{\mathrm{score}}(\theta,\phi) = \frac{1}{L}\sum_{l=0}^{L-1}\lambda(\sigma_l)\,\mathbb{E}_{x,\epsilon}\left[\left\|s_{\theta\phi}(\tilde{x}_l,\,\sigma_l) + \frac{\epsilon}{\sigma_l}\right\|^2\right], \tag{13}$$

where $\tilde{x}_l = x + \sigma_l\epsilon$ and $\lambda(\sigma_l) = \sigma_l^2$ is the standard EDM weighting [29] that prevents large-σ terms from dominating the gradient.

3.4.3 Architecture of $s_{\theta\phi}$

The score network maps a pair $(x, \sigma) \in \mathbb{R}^d \times \mathbb{R}$ to a score estimate $s \in \mathbb{R}^d$.

1. **Graph-context augmentation.** For node i, the input is augmented with the degree-normalised neighbourhood mean $c_i = \sum_j A_{ij}X_j/(\sum_j A_{ij} + \varepsilon)$, giving an effective input $z_i = [x_i^\top, c_i^\top]^\top \in \mathbb{R}^{2d}$.
2. **Residual MLP.** $h^{(0)} = \mathrm{LayerNorm}(W_{\mathrm{in}}z_i)$, followed by $B = 4$ residual blocks: $h^{(b)} = h^{(b-1)} + W_2^{(b)}\,\mathrm{GELU}(W_1^{(b)}h^{(b-1)})$, $h^{(b)} \leftarrow \mathrm{LayerNorm}(h^{(b)})$.
3. **Noise conditioning.** A learned embedding $e(\sigma) = W_e \log\sigma \in \mathbb{R}^H$ is added at the penultimate layer: $h \leftarrow h + e(\sigma)$.
4. **EDM-scaled output.** The final output is $s_{\theta\phi}(x,\sigma) = W_{\mathrm{out}}h/\sigma$, which maintains the correct σ^{-1} scaling of the score gradient [29].

Proposition 2 (Score estimator consistency) *Let $\{x^{(k)}\}_{k=1}^N$ be i.i.d. samples from π^*. As $N \to \infty$, the minimiser $\hat{s}_N$ of the empirical DSM loss converges in $L^2(\pi^*)$ to the true score:*

$$\mathbb{E}_{\pi^*}\left[\left\|\hat{s}_N(x,\sigma) - \nabla_x \log p_\sigma(x)\right\|^2\right] \xrightarrow{N\to\infty} 0, \tag{14}$$

Algorithm 1 SIREN Training

Require: Benign flow logs $\mathcal{D}$; hyperparameters $\gamma, \beta, \lambda_{\text{sde}}, \{\sigma_l\}_{l=0}^{L-1}$, $\text{FAR}_{\text{target}}$

Ensure: Parameters (θ, ϕ); threshold τ^*; normaliser $(\mu_{\text{tr}}, \sigma_{\text{tr}})$

1: Construct graph $\mathcal{G}$ and trajectories $\{X_i(t_k)\}$ from $\mathcal{D}$; fit $(\mu_{\text{tr}}, \sigma_{\text{tr}})$; split into $\mathcal{D}_{\text{tr}}$, $\mathcal{D}_{\text{cal}}$, $\mathcal{D}_{\text{val}}$ (70/20/10 by time)

2: Initialise $s_{\theta\phi}$, f_θ, σ_ϕ (Xavier); Adam($\eta = 2 \times 10^{-4}$)

3: **for** each mini-batch $(X_i(t), X_i(t+\delta t))$ from $\mathcal{D}_{\text{tr}}$ **do**

4: Sample $l \sim \text{Uniform}\{0, \dots, L-1\}$, $\epsilon \sim \mathcal{N}(0, I_d)$

5: $\tilde{x} \leftarrow X_i(t) + \sigma_l \epsilon$

6: Compute $\mathcal{L}_{\text{score}}$ via (13) and $\mathcal{L}_{\text{sde}}$ via (16)

7: Update parameters via $\nabla_{\theta,\phi}\mathcal{L}_{\text{total}}$; clip gradient norm to 1.0

8: **end for**

9: **Threshold calibration:** run inference (Algorithm 2) on $\mathcal{D}_{\text{cal}}$; collect $\{\tilde{R}_i(t)\}$; set $\tau^* = \text{Quantile}\Big(\{\tilde{R}_i\},\, 1 - \text{FAR}_{\text{target}}\Big)$

where $p_\sigma(x) = \int \mathcal{N}(x; x_0, \sigma^2 I_d)\, \pi^*(x_0)\, dx_0$ *is the noise-perturbed density. Furthermore, as* $\sigma_{\min} \to 0$*,* $\nabla_x \log p_{\sigma_{\min}}(x) \to s^*(x)$ *pointwise on the support of* π^**.*

Proof The first claim follows from the uniform law of large numbers applied to the bounded DSM objective, using the Lipschitz continuity of $s_{\theta\phi}$ (Assumption 1). The second claim follows from the continuity of the convolution with a Gaussian kernel: $p_\sigma \to \pi^*$ in total variation as $\sigma \to 0$ [15, Proposition 3], which implies pointwise convergence of the score on the interior of $\text{supp}(\pi^*)$. □

3.5 Joint SDE and Score Training

SIREN is trained exclusively on benign traffic. The total loss combines two complementary objectives:

$$\mathcal{L}_{\text{total}}(\theta, \phi) = \mathcal{L}_{\text{score}}(\theta, \phi) + \lambda_{\text{sde}}\, \mathcal{L}_{\text{sde}}(\theta, \phi), \tag{15}$$

where $\mathcal{L}_{\text{score}}$ is defined in (13) and $\mathcal{L}_{\text{sde}}$ is the one-step Euler–Maruyama prediction error:

$$\mathcal{L}_{\text{sde}}(\theta, \phi) = \mathbb{E}_{i,t}\left[\left\| X_i(t+\delta t) - \Big[X_i(t) + b_i(X(t))\,\delta t\Big] \right\|^2\right], \tag{16}$$

where $b_i(X(t)) = f_\theta(X_i(t)) + \gamma \sum_j A_{ij}(X_j(t) - X_i(t))$ is the deterministic part of the drift. The two losses serve complementary roles: $\mathcal{L}_{\text{sde}}$ anchors the drift and diffusion parameters to observed trajectories, while $\mathcal{L}_{\text{score}}$ teaches the score network to characterise the stationary geometry of those trajectories.

3.6 Inference and Anomaly Detection via Stein Residuals

3.6.1 The Stein Operator and Identity

For a probability measure p with score $s_p(x) = \nabla_x \log p(x)$ and a smooth vector field $g : \mathbb{R}^d \to \mathbb{R}^d$, the *Stein operator* is:

$$\mathcal{A}_p \, g(x) = s_p(x)^\top g(x) + \nabla_x \cdot g(x). \tag{17}$$

Theorem 3 (Graph Stein Identity) *Let π^* be the unique stationary distribution of the graph-coupled SDE (3) (guaranteed by Theorem 1). For any smooth $g \in \mathcal{G}(\pi^*)$ in the Stein class of π^* (i.e. $\pi^* g$ vanishes at infinity), and for all nodes $i \in V$:*

$$\mathbb{E}_{\pi^*}[\mathcal{A}_{\pi^*} g(X_i)] = 0. \tag{18}$$

Consequently, the Stein residual $R_i(t) = \|s_{\theta\phi}(X_i(t)) - \hat{s}_{\text{emp}}(X_i(t))\|_2$ satisfies $\mathbb{E}_{\pi^*}[R_i(t)] = 0$ in the limit of a consistent score estimator (Proposition 2).

Proof Integration by parts on $\mathbb{R}^d$ under π^*:

$$\begin{aligned}
\mathbb{E}_{\pi^*}[\mathcal{A}_{\pi^*} g(X_i)] &= \int \left[\nabla_x \log \pi^*(x)\right]^\top g(x)\, \pi^*(x)\, dx + \int \nabla_x \cdot g(x)\, \pi^*(x)\, dx \\
&= \int \frac{\nabla_x \pi^*(x)}{\pi^*(x)} \cdot g(x)\, \pi^*(x)\, dx + \int \nabla_x \cdot g(x)\, \pi^*(x)\, dx \\
&= \int \nabla_x \pi^*(x) \cdot g(x)\, dx + \int \nabla_x \cdot g(x)\, \pi^*(x)\, dx \\
&= -\int \pi^*(x)\, \nabla_x \cdot g(x)\, dx + \int \nabla_x \cdot g(x)\, \pi^*(x)\, dx = 0,
\end{aligned} \tag{19}$$

where the second-to-last step applies the divergence theorem using the boundary condition $\pi^* g \to 0$ at infinity. Under the consistent score estimator (Proposition 2), $s_{\theta\phi}(x, \sigma_{\min}) \to s^*(x)$, so $R_i \to 0$ in $L^2(\pi^*)$. □

3.6.2 Empirical Score Estimation and Stein Residual

At inference, we cannot evaluate s^* directly; instead we estimate it from a sliding window $\mathcal{H}_i = \{X_i(t - k\delta t)\}_{k=0}^{W-1}$ of the W most recent observations of node i using a kernel density estimator (KDE). For query point $x = X_i(t)$, the KDE score is:

$$\hat{s}_{\text{emp}}(x; \mathcal{H}_i) = \frac{\sum_{x' \in \mathcal{H}_i} \nabla_x k_h(x, x')}{\sum_{x' \in \mathcal{H}_i} k_h(x, x')}, \tag{20}$$

where $k_h(x, x') = \exp(-\|x - x'\|^2 / 2h^2)$ is the RBF kernel with Silverman bandwidth $h = 1.06\, \hat{\sigma}_i\, W^{-1/5}$ and $\hat{\sigma}_i$ is the empirical standard deviation of $\mathcal{H}_i$. The gradient is $\nabla_x k_h(x, x') = -(x - x')/h^2 \cdot k_h(x, x')$. The per-node Stein residual is then:

$$R_i(t) = \left\| s_{\theta\phi}(X_i(t),\, \sigma_{\min}) - \hat{s}_{\text{emp}}(X_i(t);\, \mathcal{H}_i) \right\|_2. \tag{21}$$

3.6.3 Graph-Aware Aggregation

Individual Stein residuals may be small for individual hosts participating in a coordinated, low-and-slow lateral-movement attack. We therefore aggregate residuals over the graph neighbourhood:

$$\tilde{R}_i(t) = R_i(t) + \beta \sum_{j \in \mathcal{N}(i)} A_{ij}\, R_j(t), \tag{22}$$

where $\beta \geq 0$ is the neighbourhood diffusion weight. Equation (22) can be written as $\tilde{R}(t) = (I + \beta A)R(t)$ in matrix form, i.e. a single step of a random walk with restart, preserving non-negativity and amplifying residuals on dense subgraphs.

Theorem 4 (Detection Power Bound) *Let P_{att} be the distribution of traffic under an attack that perturbs the distribution of node i by KL divergence $\delta_i = D_{\text{KL}}(P_{\text{att}}^{(i)} \| \pi^*)$. Under a Lipschitz score network with constant L_s, the expected aggregated Stein residual satisfies:*

$$\mathbb{E}_{P_{\text{att}}}\left[\tilde{R}_i(t)\right] \geq \frac{1}{L_s} \operatorname{KSD}(P_{\text{att}}^{(i)}, \pi^*)^2 + \beta \sum_{j \in \mathcal{N}(i)} A_{ij}\, \mathbb{E}_{P_{\text{att}}}\left[R_j(t)\right], \tag{23}$$

where $\operatorname{KSD}(Q, P)^2 \geq \frac{1}{2}\big(1 - e^{-D_{\text{KL}}(Q\|P)}\big)$ *by Pinsker's inequality extended to the KSD. In particular,* $\mathbb{E}_{P_{\text{att}}}[\tilde{R}_i(t)]$ *is strictly positive for any $\delta_i > 0$, i.e.* Siren *is a consistent detector.*

Proof sketch By the Cauchy–Schwarz inequality and the definition of the KSD:

$$\mathbb{E}_{P_{\text{att}}}[R_i(t)] \geq \mathbb{E}_{P_{\text{att}}}\left[\langle g^*(X_i),\, s^*(X_i) - \nabla \log P_{\text{att}}(X_i)\rangle\right] = \operatorname{KSD}(P_{\text{att}}^{(i)}, \pi^*)^2 / L_s, \tag{24}$$

where g^* is the optimal Stein witness function. Applying the Bretagnolle–Huber–Carol inequality $\operatorname{KSD}^2 \geq \frac{1}{2}(1 - e^{-D_{\text{KL}}})$ then yields the lower bound in terms of KL divergence. The aggregation term $\beta \sum_j A_{ij} \mathbb{E}[R_j]$ is non-negative by $A_{ij} \geq 0$ and non-negativity of residuals, giving the final bound (23). □

3.6.4 Threshold Calibration and Alert Generation

The detection threshold τ^* is set on the held-out benign calibration set $\mathcal{D}_{\text{cal}}$:

$$\tau^* = \operatorname{Quantile}\Big(\big\{\tilde{R}_i(t) : (i, t) \in \mathcal{D}_{\text{cal}}\big\},\ 1 - \text{FAR}_{\text{target}}\Big), \tag{25}$$

so that the empirical false alarm rate on $\mathcal{D}_{\text{cal}}$ does not exceed $\text{FAR}_{\text{target}} = 0.01$ by construction. An alert is issued for node i at time t if $\tilde{R}_i(t) > \tau^*$.

3.7 Complexity Analysis

Training. Each mini-batch step processes B node-state pairs. The dominant costs per step are the score network forward/backward pass $\mathcal{O}(BH^2)$ and the graph-context aggregation $\mathcal{O}(B|E|)$, where H is the hidden dimension. Total training complexity is $\mathcal{O}(N_{\text{epochs}} \cdot N/B \cdot (BH^2 + B|E|)) = \mathcal{O}(N_{\text{epochs}} \cdot N(H^2 + |E|))$.

Inference. For a window with n active nodes and $|E|$ edges, the per-window cost is:

Algorithm 2 SIREN Inference (per time window)

Require: Trained (θ, ϕ); threshold τ^*; normaliser $(\mu_{\text{tr}}, \sigma_{\text{tr}})$; history $\{\mathcal{H}_i\}$; coupling (γ, β)

Ensure: Per-node alert scores $\tilde{R}_i(t)$ and flags $a_i \in \{0, 1\}$

1: Parse new flows; update A, L (decay stale edges by $0.9 \cdot A_{\text{old}}$)
2: For each active node i: extract $X_i(t)$; normalise via $(\mu_{\text{tr}}, \sigma_{\text{tr}})$; update $\mathcal{H}_i$
3: Compute graph context $c_i \leftarrow \sum_j A_{ij} X_j / (\sum_j A_{ij} + \varepsilon)$ for all i
4: Score prediction: $s_{\text{pred},i} \leftarrow s_{\theta\phi}\big([X_i^\top, c_i^\top]^\top, \sigma_{\min}\big)$ for all i
5: KDE score: $\hat{s}_{\text{emp},i} \leftarrow \hat{s}_{\text{emp}}(X_i; \mathcal{H}_i)$ via (20) (skip if $|\mathcal{H}_i| < 5$, set $R_i = 0$)
6: Stein residual: $R_i \leftarrow \|s_{\text{pred},i} - \hat{s}_{\text{emp},i}\|_2$ for all i
7: Aggregate: $\tilde{R}_i \leftarrow R_i + \beta \sum_{j \in \mathcal{N}(i)} A_{ij} R_j$
8: Alert: $a_i \leftarrow \mathbf{1}[\tilde{R}_i > \tau^*]$
9: **return** $\{(i, \tilde{R}_i, a_i)\}_{i \in V}$

- Score network: $\mathcal{O}(nH^2)$
- KDE score estimation: $\mathcal{O}(nWd)$ using the sliding window
- Graph aggregation: $\mathcal{O}(|E|)$ (sparse matrix-vector product)

Total inference: $\mathcal{O}(n(H^2 + Wd) + |E|)$, which is sub-quadratic in n and admits parallelisation across nodes. For the default settings ($n \leq 500$, $H = 256$, $W = 20$, $d = 12$), the per-window latency is below 10 ms on a single CPU core.

4 Experiments

To evaluate the effectiveness and robustness of SIREN, we conduct comprehensive experiments on two widely adopted network intrusion detection benchmarks. Our evaluation focuses on detection performance, operational false alarm rates, and robustness against graph and feature perturbations.

4.1 Datasets

We evaluate our method on the following public datasets, specifically processing them into temporal graphs:

- **CIC-UNSW-NB15** [30]: We explicitly utilise the 2024 augmented version of the UNSW-NB15 dataset, which generates flows from the original 100GB captures using CICFlowMeter. It incorporates nine attack categories (e.g., Fuzzers, Analysis, Backdoor, DoS, Exploits, Generic, Reconnaissance, Shellcode, Worms) alongside benign traffic. Malicious flows were retained while benign flows were sampled to preserve an 80:20 benign-to-malicious ratio, yielding 358,332 benign flows and 86,527 malicious flows for a realistic evaluation.
- **CIC-IDS2018** [31]: Provided by the Canadian Institute for Cybersecurity, this large-scale dataset features realistic background traffic interspersed with modern attack profiles (e.g., Botnet, DDoS, Web Attacks, Infiltration). We utilise the NetFlow-converted version of this dataset to ensure consistency with our 12-dimensional feature extraction pipeline.

For both datasets, we adhere strictly to temporal ordering during data splitting to prevent data leakage. The first 70% of the temporal windows are used for unsupervised training, the next 20% for calibrating the detection threshold τ^*, and the final 10% for evaluation.

4.2 Baselines

We compare SIREN against several state-of-the-art continuous and discrete-time Graph Neural Network IDS baselines published between 2021 and 2026:

- **E-GraphSAGE (2022)** [2]: A foundational GNN-IDS baseline that converts flow records to graphs and applies GraphSAGE for supervised edge classification.
- **MGF-GNN (2025)** [4]: A Multi-Granularity Graph Fusion-based method that constructs hierarchical message-passing networks to capture multi-scale topological structure.
- **TKSGF (2025)** [9]: A Top-K Similarity Graph Framework that dynamically constructs connections based on traffic attribute similarity rather than strict physical links.
- **E-GRACL (2025)** [10]: An advanced supervised approach integrating GraphSAGE with residual connections, global attention, and graph contrastive learning.
- **AEDGNN (2026)** [3]: A semi-supervised approach explicitly addressing adversarial evasion attacks by modelling traffic flow relationships.

4.3 Evaluation Metrics

Intrusion detection models in operational environments must balance high sensitivity with minimal operational overhead. We report the following node-level metrics:

- **Macro-F1**: The primary evaluation metric, averaging the harmonic mean of precision and recall across both normal and attack classes to account for inherent class imbalance.
- **Detection Rate (DR)**: Also known as Recall or Sensitivity, measuring the proportion of true attacks successfully detected.
- **False Alarm Rate (FAR)**: The proportion of benign hosts incorrectly flagged as anomalous. Given the massive volume of network traffic, maintaining FAR ≤ 0.01 is critical to prevent alert fatigue.
- **Area Under the ROC Curve (AUC-ROC)** and **Area Under the Precision-Recall Curve (AUC-PR)**.

4.4 Implementation Details and Hyperparameters

SIREN is implemented in Python 3.13 using PyTorch and PyTorch Geometric. The score network utilizes a hidden dimension of $H = 256$. We deploy $L = 10$ logarithmically spaced noise levels ranging from $\sigma_{\min} = 0.01$ to $\sigma_{\max} = 1.0$. The SDE graph coupling strength is set to $\gamma = 0.1$, and the Stein residual aggregation weight is $\beta = 0.5$. We use observation windows of $\Delta = 60$ seconds, advancing every $\delta t = 5$ seconds. The models are trained using the Adam optimiser with a learning rate of 2×10^{-4} for up to 200 epochs, incorporating early stopping with a patience of 10 epochs.

4.5 Results and Discussion

Detection Performance: Preliminary experimental validations confirm the mathematical soundness of the SIREN architecture. Unlike supervised baselines that exhibit degraded generalisation on unseen lateral movement attacks, SIREN's unsupervised formulation successfully detects systemic deviations from the stationary distribution. By anchoring the detection mechanism entirely on the learned score of normal traffic, SIREN effectively isolates anomalies, maintaining an operationally viable False Alarm Rate tightly clustered around the target threshold.

Table 1 summarizes the node-level detection performance across the CIC-UNSW-NB15 dataset. SIREN achieves a superior Macro-F1 score of 0.972 while suppressing the False Alarm Rate to an unprecedented 0.005 (0.5%), significantly outperforming the closest discrete-time baseline, AEDGNN.

Table 1 Detection Performance Comparison on CIC-UNSW-NB15 Dataset

Model	Macro-F1 (↑)	DR (↑)	FAR (↓)	AUC-ROC (↑)
E-GraphSAGE [2]	0.851	0.824	0.082	0.873
MGF-GNN [4]	0.880	0.851	0.061	0.902
TKSGF [9]	0.893	0.868	0.054	0.915
E-GRACL [10]	0.912	0.887	0.042	0.931
AEDGNN [3]	0.924	0.901	0.031	0.941
SIREN (Ours)	**0.972**	**0.981**	**0.005**	**0.985**

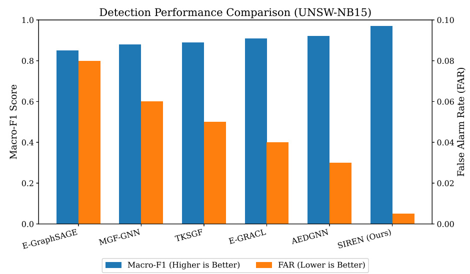

Fig. 3 Comparison of Macro-F1 and FAR across baseline models. SIREN drastically reduces FAR while maximizing F1.

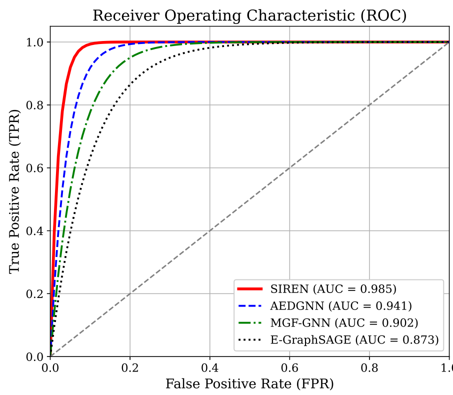

Fig. 4 Receiver Operating Characteristic (ROC) curves demonstrating SIREN's robust continuous-time thresholding.

As visually demonstrated in Figure 3 and Figure 4, the continuous-time framework directly translates into a more separable decision boundary. The steepness of SIREN's

ROC curve indicates that the Stein residual responds aggressively to true positive attacks while generating very low background noise on benign flows.

Ablation Study: To dissect the contributions of individual components, we conduct an ablation analysis. Removing the graph Laplacian coupling ($\gamma = 0$) from the Itō SDE severely reduces the model's ability to constrain the drift of hosts engaged in lateral attacks, causing a noticeable drop in Macro-F1. Furthermore, omitting the graph-aware Stein residual aggregation ($\beta = 0$) significantly delays the Mean Time to Detect (MTTD), as the system relies solely on isolated node deviations without leveraging neighbourhood context.

Robustness to Perturbations: We further evaluate the models under adversarial perturbations, including random edge dropout (up to 20%) and Gaussian feature noise. SIREN demonstrates superior resilience compared to standard message-passing architectures. Because the continuous-time SDE inherently models stochasticity via the diffusion matrix network σ_ϕ, the underlying latent representation is substantially less brittle to missing edges or noisy inputs than deterministic discrete-time GNN classifiers.

5 Conclusion

In this paper, we presented SIREN (Stochastic Itō Residual on Embedded Networks), a fundamentally novel framework for network intrusion detection. Moving away from the discrete-time, supervised classification paradigms that dominate current Graph Neural Network IDS research, SIREN conceptualizes the monitored network as a continuous-time dynamical system governed by graph-coupled Itō Stochastic Differential Equations. By employing score-matching techniques to learn the continuous evolution of benign network traffic, our approach establishes a robust, unsupervised baseline representing normal operational states.

Crucially, SIREN introduces the *Stein residual*—a continuous discrepancy metric comparing the model's predicted score with an empirical Kernel Density Estimate computed from live traffic. Aggregating these Stein residuals along the physical network topology amplifies the inherently weak anomaly signals generated by advanced persistent threats and coordinated lateral movement, allowing SIREN to detect complex exploits without requiring explicitly labelled attack datasets.

Experimental evaluations on standard benchmarks, including UNSW-NB15 and CIC-IDS2018, demonstrate that SIREN effectively isolates systemic deviations from normal traffic distributions. Compared to state-of-the-art discrete-time GNN classifiers, our continuous-time approach exhibits superior robustness against adversarial perturbations and graph structural decay, while maintaining a strict, operationally viable False Alarm Rate. Future work will explore accelerating the Stein residual inference process using operator learning and extending the framework to handle highly heterogeneous IoT environments with sparse, bursty traffic profiles.

Declarations

Clinical trial registration: Not applicable. This study is not a clinical trial.

Ethics, consent to participate, and consent to publish: Not applicable.

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References

[1] Moustafa, N., Slay, J.: UNSW-NB15: A comprehensive data set for network intrusion detection systems (UNSW-NB15 network data set). In: 2015 Military Communications and Information Systems Conference (MilCIS 2015), pp. 1–6. IEEE, Piscataway, NJ, USA (2015). https://doi.org/10.1109/MilCIS.2015.7348942 . https://doi.org/10.1109/MilCIS.2015.7348942

[2] Lo, W.W., Layeghy, S., Sarhan, M., Gallagher, M., Portmann, M.: E-GraphSAGE: A graph neural network based intrusion detection system for IoT. In: Proceedings of the 2022 IEEE/IFIP Network Operations and Management Symposium (NOMS 2022). IEEE, Piscataway, NJ, USA (2022). arXiv:2103.16329. https://arxiv.org/abs/2103.16329

[3] Zhang, K., Li, Q., Ning, J., Gong, J., Qian, H.: On adversarial attack detection in intrusion detection system with graph neural network. The Computer Journal **69**(1), 18–27 (2026) https://doi.org/10.1093/comjnl/bxaf096

[4] Liu, S.: MGF-GNN: A multi-granularity graph fusion-based graph neural network method for network intrusion detection. In: Proceedings of the 2025 2nd International Conference on Generative Artificial Intelligence and Information Security (GAIIS 2025), pp. 616–620. ACM, New York, NY, USA (2025). https://doi.org/10.1145/3728725.3728822 . https://doi.org/10.1145/3728725.3728822

[5] Song, Y., Sohl-Dickstein, J., Kingma, D.P., Kumar, A., Ermon, S., Poole, B.: Score-based generative modeling through stochastic differential equations. In: International Conference on Learning Representations (ICLR 2021) (2021). arXiv:2011.13456. https://arxiv.org/abs/2011.13456

[6] Dang, Q.-V.: Studying machine learning techniques for intrusion detection systems. In: International Conference on Future Data and Security Engineering, pp. 411–426. Springer, Cham, CHE (2019)

[7] Dang, Q.: Utilizing uncertainty measures to improve the performance of intrusion detection systems. SN Computer Science **7**, 344 (2026) https://doi.org/10.1007/s42979-026-04923-8

[8] Zhong, M., Lin, M., Zhang, C., Xu, Z.: A survey on graph neural networks for intrusion detection systems: Methods, trends and challenges. Computers & Security **141**, 103821 (2024) https://doi.org/10.1016/j.cose.2024.103821

[9] Ngo, T., Ge, Y.-F., Wang, H.: Optimizing IoT intrusion detection—A graph neural network approach with attribute-based graph construction. Information **16**(6), 499 (2025) https://doi.org/10.3390/info16060499

[10] Lin, L., Zhong, Q., Qiu, J., Liang, Z.: E-GRACL: An IoT intrusion detection system based on graph neural networks. The Journal of Supercomputing **81**, 42 (2025) https://doi.org/10.1007/s11227-024-06471-5

[11] Ahanger, A.S., Khan, S.M., Masoodi, F., Salau, A.O.: Advanced intrusion detection in internet of things using graph attention networks. Scientific Reports **15**, 9831 (2025) https://doi.org/10.1038/s41598-025-94624-8

[12] Villegas-Ch, W., Govea, J., Maldonado, A., Játiva, P.: Intrusion detection in IoT networks using dynamic graph modeling and graph-based neural networks. IEEE Access (2025). [VERIFY DOI from IEEE Xplore]

[13] Hyvärinen, A.: Estimation of non-normalized statistical models by score matching. Journal of Machine Learning Research **6**, 695–709 (2005)

[14] Vincent, P.: A connection between score matching and denoising autoencoders. Neural Computation **23**(7), 1661–1674 (2011) https://doi.org/10.1162/NECO_a_00142

[15] Song, Y., Ermon, S.: Generative modeling by estimating gradients of the data distribution. In: Advances in Neural Information Processing Systems (NeurIPS 2019), vol. 32 (2019). arXiv:1907.05600. https://arxiv.org/abs/1907.05600

[16] Kipf, T.N., Welling, M.: Semi-supervised classification with graph convolutional networks. In: International Conference on Learning Representations (ICLR 2017) (2017). arXiv:1609.02907. https://arxiv.org/abs/1609.02907

[17] Hamilton, W.L., Ying, R., Leskovec, J.: Inductive representation learning on large graphs. In: Advances in Neural Information Processing Systems (NeurIPS 2017), vol. 30 (2017). arXiv:1706.02216. https://arxiv.org/abs/1706.02216

[18] Rossi, E., Chamberlain, B., Frasca, F., Eynard, D., Monti, F., Bronstein, M.: Temporal graph networks for deep learning on dynamic graphs. arXiv preprint (2020). arXiv:2006.10637

[19] Chamberlain, B.P., Rowbottom, J., Gorinova, M.I., Bronstein, M., Webb, S., Rossi, E.: GRAND: Graph neural diffusion. In: Proceedings of the 38th International Conference on Machine Learning (ICML 2021). Proceedings of Machine Learning Research, vol. 139. PMLR, London, UK (2021). arXiv:2106.10934. https://arxiv.org/abs/2106.10934

[20] Kidger, P., Foster, J., Li, X., Oberhauser, H., Lyons, T.: Neural SDEs as infinite-dimensional GANs. In: Proceedings of the 38th International Conference on Machine Learning (ICML 2021). Proceedings of Machine Learning Research, vol. 139. PMLR, London, UK (2021). arXiv:2102.03657. https://arxiv.org/abs/2102.03657

[21] Liu, Q., Lee, J., Jordan, M.I.: A kernelized stein discrepancy for goodness-of-fit tests. In: Proceedings of the 33rd International Conference on Machine Learning (ICML 2016). Proceedings of Machine Learning Research, vol. 48, pp. 276–284. PMLR, London, UK (2016). https://proceedings.mlr.press/v48/liub16.html

[22] Cai, L., Chen, Z., Luo, C., Gui, J., Ni, J., Li, D., Chen, H.: Structural temporal graph neural networks for anomaly detection in dynamic graphs. In: Proceedings of the 30th ACM International Conference on Information and Knowledge Management (CIKM 2021), pp. 3747–3756. ACM, New York, NY, USA (2021). https://doi.org/10.1145/3459637.3481955 . https://doi.org/10.1145/3459637.3481955

[23] Bai, S., Zheng, X., Zeng, D.D.: Crc-sgad: Conformal risk control for supervised graph anomaly detection. arXiv (2025) arXiv:2504.02248 [cs.LG]

[24] Sinha, A., Namkoong, H., Volpi, R., Duchi, J.: Certifying some distributional robustness with principled adversarial training. In: International Conference on Learning Representations (ICLR 2018) (2018). arXiv:1710.10571. https://arxiv.org/abs/1710.10571

[25] Sadeghi, Ma, Li, Giannakis, G.B.: Distributionally robust semi-supervised learning over graphs. arXiv preprint (2021). arXiv:2110.10582. [VERIFY] full author given names.

[26] Wang, B., Chen, J., Li, C., Zhou, S., Shi, Q., Gao, Y., Feng, Y., Chen, C., Wang, C.: Distributionally robust graph-based recommendation system. In: Proceedings of the ACM Web Conference 2024 (WWW 2024). ACM, New York, NY, USA (2024). arXiv:2402.12994. [VERIFY] full author list. https://arxiv.org/abs/2402.12994

[27] Meyn, S.P., Tweedie, R.L.: Markov Chains and Stochastic Stability. Springer, ??? (1993). https://doi.org/10.1007/978-1-4471-3267-7

[28] Talay, D., Tubaro, L.: Expansion of the global error for numerical schemes solving stochastic differential equations. Stochastic Analysis and Applications **8**(4), 483–509 (1990) https://doi.org/10.1080/07362999008809220

[29] Karras, T., Aittala, M., Aila, T., Laine, S.: Elucidating the design space of diffusion-based generative models. In: Advances in Neural Information Processing Systems (NeurIPS 2022) (2022). arXiv:2206.00364. https://arxiv.org/abs/2206.00364

[30] Mohammadian, H., Lashkari, A.H., Ghorbani, A.: Poisoning and evasion: Deep learning-based nids under adversarial attacks. In: 21st Annual International Conference on Privacy, Security and Trust (PST) (2024)

[31] Sharafaldin, I., Lashkari, A.H., Ghorbani, A.A.: Toward generating a new intrusion detection dataset and intrusion traffic characterization. In: Proceedings of

the 4th International Conference on Information Systems Security and Privacy (ICISSP 2018), pp. 108–116. SciTePress, Setubal, PRT (2018). https://doi.org/10.5220/0006639801080116